

PYTHON 10 PROGRAMS

Problem 1 – Multiplication Table

Ask for a number and print its table from 1 to 10. Focus: for + range().

```
num=int(input("enter a number:"))
for i in range(1,11):
    print(f"{num} * {i} ={num*i}")
```

output:

enter a number:5

5 * 1 =5

5 * 2 =10

5 * 3 =15

5 * 4 =20

5 * 5 =25

5 * 6 =30

5 * 7 =35

5 * 8 =40

5 * 9 =45

5 * 10 =50

Problem 2 – Sum of Numbers

Ask for n and calculate 1 + 2 + ... + n. Focus: accumulator + for.

```
n=int(input("enter a number:"))
sum=0
for i in range(n+1):
    sum+=i
print(sum)
```

output:

enter a number:6

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Problem 3 – Count Even and Odd Numbers

For 1 to n, count even and odd numbers. Focus: for + if.

```

n=int(input("enter a number:"))
even=0
odd=0
for i in range(1,n+1):
    if i %2==0:
        even+=1
    else:
        odd+=1
print(f"the odd numbers are : {odd}")
print(f"the even numbers are : {even}")

```

output:

enter a number:6

the odd numbers are : 3

the even numbers are : 3

Problem 4 – Prime Number Checker

Determine whether a number is prime. First explain what makes a number prime.

Ans:

If a numbers is said to prime if it divisible by only 1 and itself.

```

num=int(input("enter number:"))
count=0
for i in range(1,num+1):
    if num%i==0:
        count+=1
    else:
        count+=0
if count==1:
    print(f"{num} is prime")
else:
    print(f"{num} is not a prime")

```

output:

enter number:9

9 is not a prime

Problem 5 – Palindrome Number

Check whether a number such as 121 is a palindrome. Try without a ready-made palindrome function.

Ans: a number is said to be palindrome if it equals to reverse of the same number.

```

num=int(input("enter a number:"))
reverse=0
while num>0:

```

```

        number=num%10
        reverse=reverse*10+number
        num=num//10
    if reverse==num:
        print("it is a plindrome")
    else:
        print("it is not a palindrome")

```

test case1:

enter a number:121

it is a palindrome

test case 2:

enter a number:123

it is not a palindrome

Problem 6 – Find Factors

Print all factors of a number. Example: 12 → 1, 2, 3, 4, 6, 12.

```

num=int(input("enter number:"))
for i in range(1,num+1):
    if num%i==0:
        print(i,end=" ")

```

test case ;

enter number:12

1 2 3 4 6 12

Problem 7 – break Challenge

Search 1–100 and stop at the first number divisible by both 7 and 9. Use break.

```

for i in range(1,100):
    if i%7==0 and i%9==0:
        print(i)
        break

```

output: 63

Problem 8 – continue Challenge

Print 1–50 but skip numbers divisible by 3. Use continue.

```

for i in range(1,50):
    if i%3==0 :
        continue
    print(i)

```

output:

1,2,4,5,7,8,10,11,13,14,16,17,19,20,22,23,25,26,28,29,31,32,34,35,37,38,40,41,43,44,46,47,49.

Problem 9 – Pattern Printing

Print *, **, ***, ****, ***** and the reverse. Bonus: centered pyramid.

```
for i in range(6):
    print(i*" ",end=" ")
    print()
    i+=1
```

output:

*

**

reversed:

```
for i in range(5,0,-1):
    print(i*" ",end=" ")
    print()
```

output:

**

*

Problem 10 – Number Pattern

Print 1, 12, 123, 1234, 12345 and then 1, 22, 333, 4444, 55555. Use nested loops.

```
for i in range(1, 6):
    for j in range(1, i + 1):
        print(j, end="")
    print()
```

output:

1

12

123

1234

12345

```
for i in range(1, 6):  
    for j in range(i):  
        print(i, end="")  
    print()
```

output:

1

22

333

4444

55555